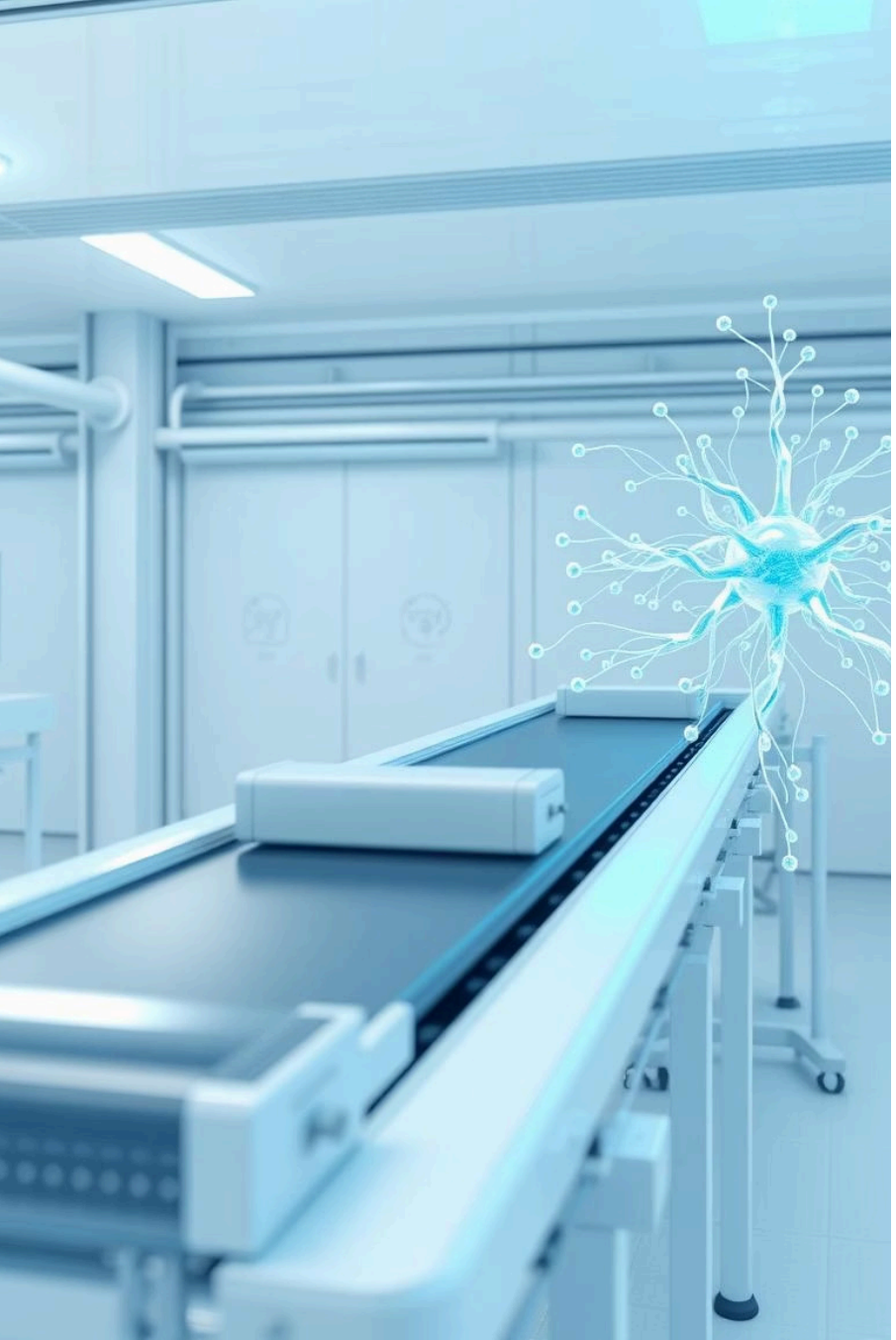

AI Agents
Introduction & Anatomy

From fixed workflows to adaptive reasoning

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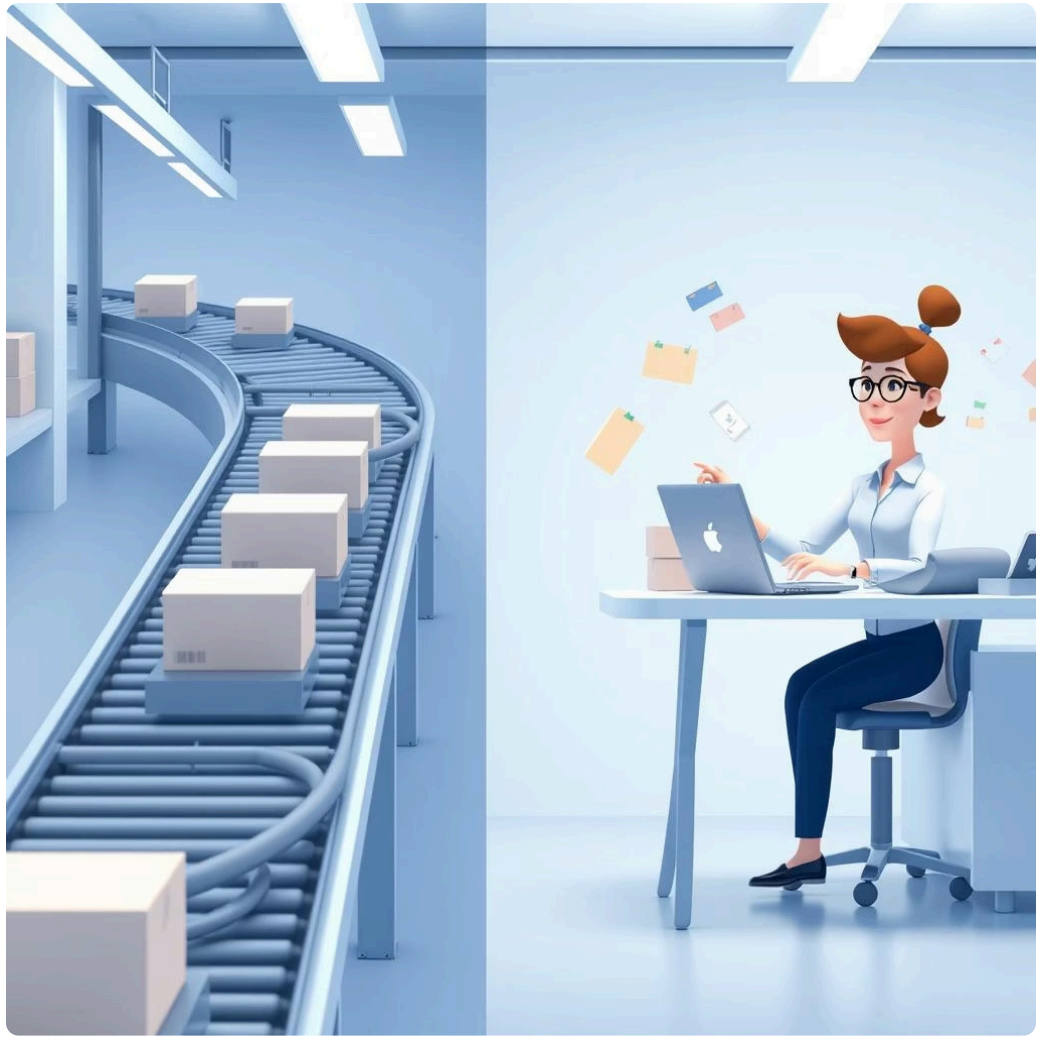
Conveyor Belt vs. Employees

Imagine your automation as either a conveyor belt or an employee.

- A conveyor belt: deterministic, rigid, always does the same thing.
- An employee: adapts, reasons, and uses the right tools for the task.

> **Workflows = "Do exactly as I said."**
>
> **Agents = "Do what makes sense to achieve the goal."**



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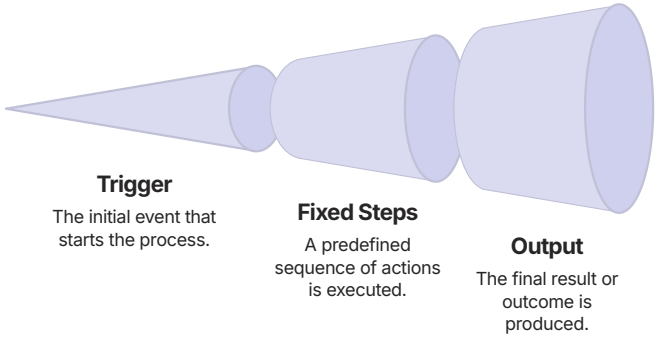
Comparing Workflows & Agents

1

Workflows

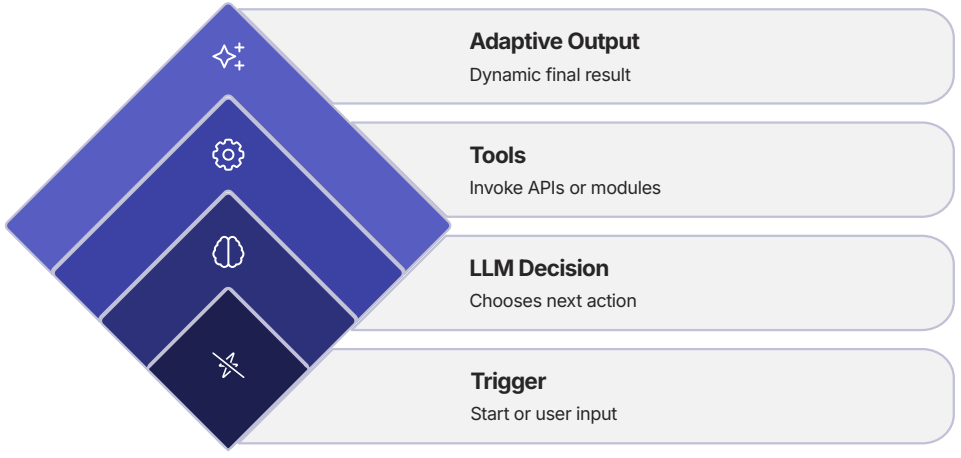
Trigger → fixed steps → output. Excels at repetitive tasks where input is predictable.



2

Agents

Trigger → LLM decides → picks tools → adaptive output. Excels when input is unpredictable.



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Workflow vs Agent Demo

Let's consider a practical example. We'll compare how a workflow and an AI agent handle the same request:

Traditional Workflow

- Webhook receives data.
- Data is directly added to a Google Sheet.
- A generic email is sent via Gmail.

AI Agent Approach

- Webhook receives data.
- AI Agent interprets the request.
- Agent dynamically decides to use Google Sheets and/or Gmail based on need.

The key difference: rigid execution versus adaptive problem-solving to achieve the goal.

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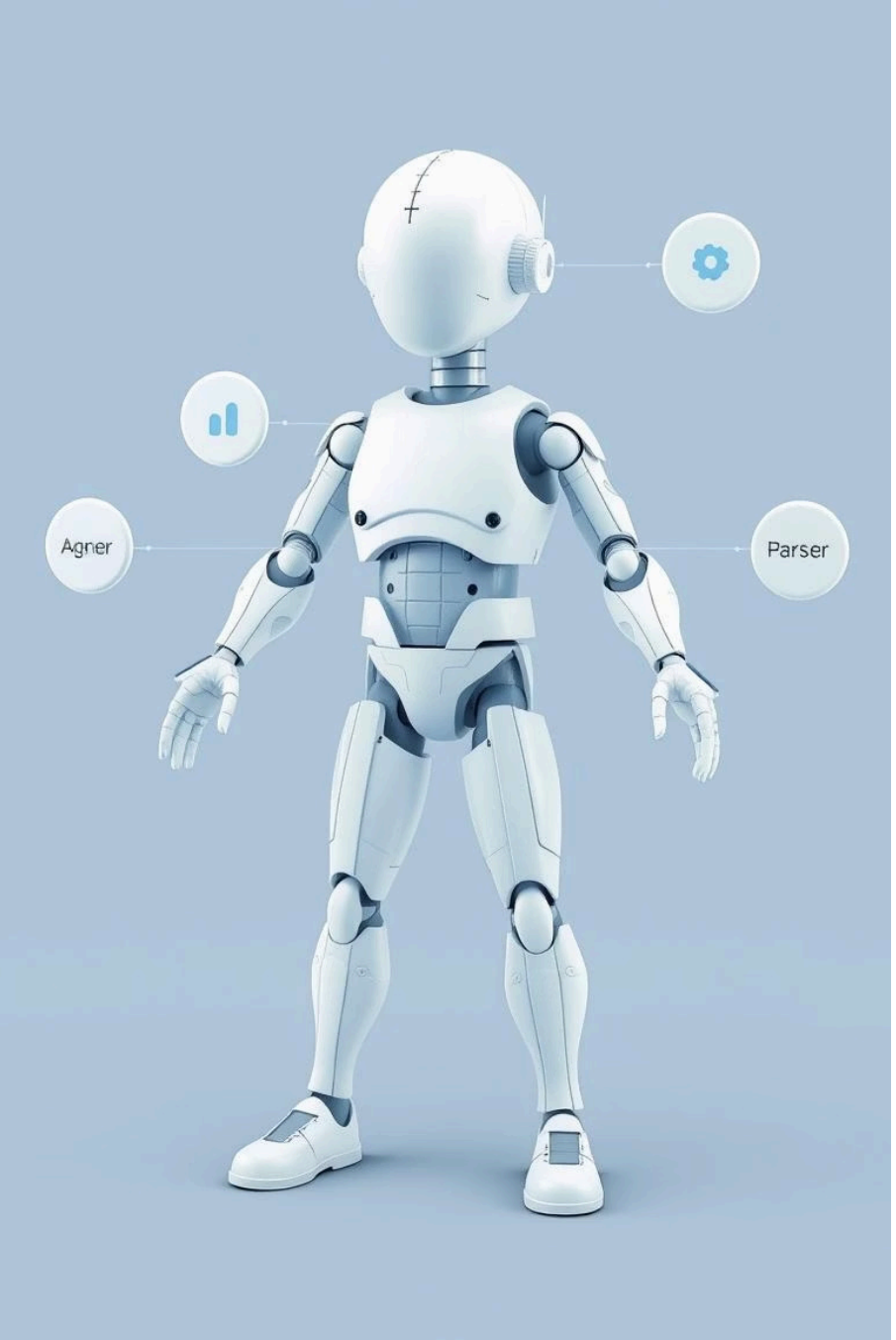

Agent Anatomy — Core Components

Just like a human, an AI agent has several interconnected parts that enable it to understand, reason, and act effectively:

- 🧠 **Brain (Model):** The core intelligence.
- 📝 **Memory:** Retains context from past interactions.
- 🔧 **Tools:** The actions it can perform.
- 💬 **Prompts:** Its guiding instructions.
- 📊 **Output Parser:** Structures its responses.

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🧠 The Brain — LLM Model

The Large Language Model (LLM) is the central intelligence of the agent. It's responsible for:

- Reasoning through complex problems.
- Making decisions based on input and available tools.
- Generating coherent and contextually relevant responses.

Your choice of model significantly impacts the agent's capabilities, performance, and operational cost.

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📝 Memory — Context Retention

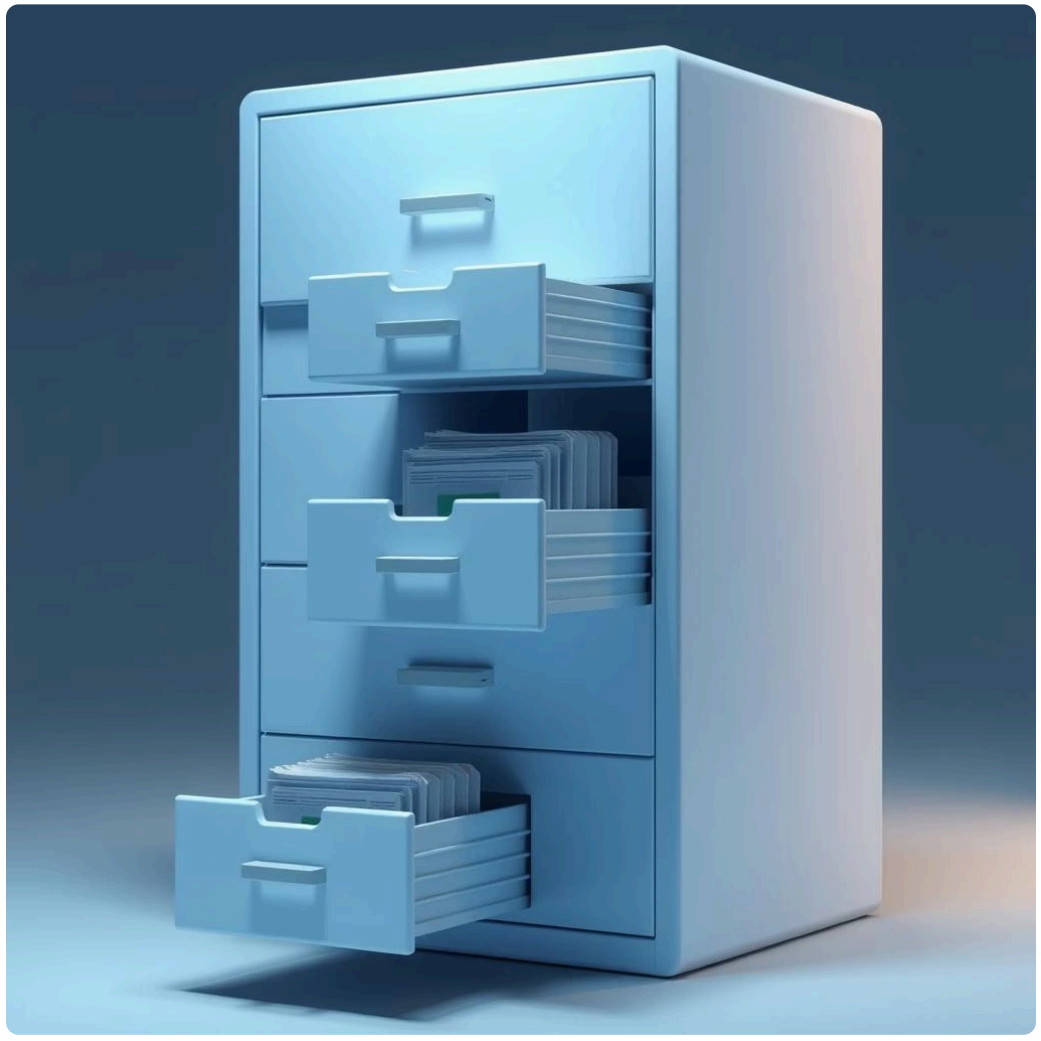
Memory allows agents to maintain context across interactions, leading to more natural and effective conversations. Without it, every interaction would be a fresh start.

Window Buffer (Short-Term)

- Remembers the last 'N' turns or messages.
- Ideal for immediate conversational flow.
- Similar to sticky notes on your desk.

Persistent Memory (Long-Term)

- Stores information in a database (e.g., DB, Redis).
- Crucial for recalling past data, user preferences, or historical context.
- Like a well-organized filing cabinet.



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🔧 Tools — The Agent’s Capabilities

Tools are the agent's action nodes—the specific integrations and functionalities it can invoke. These can range from simple data manipulation to complex API calls.

- Each tool needs a clear description so the agent knows when and how to use it.
- Well-defined tools expand the agent's ability to interact with the real world.

Example Tool Description:

"Use this tool to look up or update inventory levels in our external database."

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💬 Prompts — The Agent’s Instructions

Prompts are the guiding instructions that define an agent's persona, goals, and operational constraints. They shape how the agent interprets tasks and interacts with users.

- A well-crafted system message sets the agent's role and tone.
- They dictate how the agent should prioritize objectives and utilize its tools.

ⓘ **Example Prompt:**

"You are an attentive inventory manager for a retail chain. Always check Airtable for current stock before answering any customer query about product availability."

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📊 Output Parser — Structured Responses

The Output Parser is critical for ensuring the agent's replies are consistent and usable by downstream systems. It mandates a specific format, preventing vague or free-form responses.

- Forces the agent to output in a required format (e.g., JSON, plain text).
- Prevents rambling, inconsistent answers, or hallucinated data structures.
- Essential for integrating agent outputs into other automated processes.

Example:

Force output as: { "product": "", "quantity": "" }

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Demo: Inventory Agent Skeleton

Let's assemble an AI Agent for a practical inventory management task, showcasing each core component in action.

1. Entry Point: Webhook

Data comes in from a webhook, triggering the agent. This could be a new order or a stock update.

2. Brain: GPT-4

The powerful GPT-4 model serves as the agent's brain, interpreting the webhook data and deciding the next steps.

3. Memory: Window Buffer

A short-term window buffer retains the last 3 turns of conversation or interaction, providing immediate context.

4. Tool: Airtable

The agent uses an Airtable integration to look up or update inventory levels. Its description guides appropriate use.

5. Prompt: "Inventory Manager"

The agent's persona is defined: "You help manage product inventory," guiding its decision-making and responses.

6. Output: JSON Result

The final output is consistently structured as a JSON result, ensuring compatibility with downstream systems.

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Key Takeaways — Agents vs Workflows

Workflows: Static & Predictable

- Function like **static conveyor belts**, executing fixed steps.
- Ideal for repetitive tasks with predictable inputs.
- Output is a direct result of predefined actions.

Agents: Adaptive & Intelligent

- Act like **adaptive employees** with a brain, memory, and tools.
- Utilize clear tool descriptions and output parsers for reliable results.
- Reason, make decisions, and adapt to unpredictable inputs.
- Their ability to "decide" based on goals is key to their power.

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